Ancilla-Efficient QSample Preparation for Reversible Markov Chains

Nicholas Zhao
Department of Electrical and Electronic Engineering
Imperial College London
London, United Kingdom
nz422@ic.ac.uk

***Abstract*—Preparing quantum samples (QSAMPLES)—coherent encodings of stationary distributions of reversible Markov chains—is a fundamental primitive in quantum sampling, particularly for quantum simulated annealing. A central limitation of existing phase-estimation-based frameworks is the ancilla qubit overhead. In this work, we present a new end-to-end framework requiring only one ancilla qubit in the working register. The key technical ingredient is a selective phase compiler circuit using one ancilla qubit, built from a generalized quantum signal processing (GQSP)-based projector onto the 1-eigenspace of the qubitized Szegedy walk. Embedding these selective phase compilers into the fixed-point amplitude amplification (FPAA) procedure and iterating yields a quantum algorithm that, given an initial state, oracle access, lower bounds on the overlaps between adjacent states, and lower bounds on the phase gaps, outputs a QSAMPLE within any desired trace distance (and thus total variation distance). The query complexity scales inversely with the square roots of both the minimum overlap and the minimum spectral gap of the Markov chains across the cooling schedule, up to polylogarithmic factors. We also perform simulations to verify how our qubit and query complexity evolve with the trace distance, and how this work compares to the previous framework. These results establish two improvements over the previous framework by Wocjan and Abeyesinghe. First, the working-register ancilla cost is reduced to one. Second, by inserting our GQSP-based selective phase compiler into the FPAA procedure, we improve the QSAMPLE transport overlap dependence from $1/p_{\min}$ to $1/\sqrt{p_{\min}}$, relative to their Grover's-$\pi/3$-fixed-point. Finally, as a direct application we run the quantum algorithm to prepare a Gibbs QSAMPLE and obtain a rigorous complexity analysis.**

***Index Terms*—Markov chains, QSAMPLE, Quantum simulated annealing, Quantum walk, Fixed-point amplitude amplification, Generalized quantum signal processing**

I. INTRODUCTION

Many quantum sampling and estimation algorithms use quantum encodings of the probability distributions as input states. This is due to the amount of rich information encoded across the superposition of the coherent quantum sample, which quantum algorithms composed of unitaries can perform useful manipulations on. Let π be a probability distribution on Ω, then a QSAMPLE is a quantum state whose amplitudes are square roots of the target distribution

$$|\pi\rangle = \sum_{x\in\Omega} \sqrt{\pi(x)}\,|x\rangle\,. \tag{1}$$

Measuring $|\pi\rangle$ in the computational basis $\Pi_x := |x\rangle\langle x|$ outputs a classical sample from $\pi(x)$, while an undisturbed fully coherent quantum state $|\pi\rangle$ allows the probability distributions to be used inside quantum procedures such as amplitude amplification [1], amplitude estimation [2], and mean estimation [3]. Notably, Wocjan et al. [4] proposed a quantum sampling framework which is based on an iteration over the framework put forward by Somma et al. [5] termed Quantum Simulated Annealing (QSA). The framework constructs the Szegedy quantum walk operator [6] associated with a reversible Markov chain on two registers $\mathbb{C}^{|\Omega|}\otimes\mathbb{C}^{|\Omega|}$, and uses quantum phase estimation (QPE) to implement a reflection across the quantum stationary state $|\pi\rangle$ (also in [8]). Since this operation is governed by the phase gap $\Delta = \Omega(\sqrt{\delta(\mathcal{P})})$, where $\delta(\mathcal{P})$ is the spectral gap of the underlying Markov chain, thus the reflection cost scales as $\widetilde{\mathcal{O}}(1/\sqrt{\delta(\mathcal{P})})$. These reflections are then used to anneal through a sequence of Markov chains $\mathcal{P}_0,\ldots,\mathcal{P}_\ell$ until preparing the target QSAMPLE $|\pi_\ell\rangle$ within trace distance ε. Even though a general quantum speedup for Markov chain Monte Carlo (MCMC) methods has been conjectured to be computationally infeasible due to $\mathsf{SZK}\subseteq\mathsf{BQP}$ [29], [30]. Despite this the framework of [4] still enjoyed much success and wide adoption, particularly in research relating to partition function estimation [2], [9], [10], [24], Gibbs state preparation [11], and Bayesian inference [12]. However, this framework suffers from two shortcomings. Firstly, the number of ancilla qubits required for a QPE-based reflection scales as $\mathcal{O}(\log(1/\sqrt{\delta(\mathcal{P})})\log(\ell/\varepsilon))$. Secondly, the Grover-$\pi/3$ fixed-point search incurs a quadratically larger overlap dependence $1/p_i$ whereas the Grover optimum $1/\sqrt{p_i}$ [13] can be achieved by using fixed-point amplitude amplification (FPAA) [14]. Recent developments in generalized quantum signal processing (GQSP) [16] and eigenspace-reflection algorithms [17] also provided us with tools for implementing polynomial transformation on our desired unitaries with reduced ancilla qubit overhead.

In this work, we seek to address these shortcomings by proposing a more ancilla-efficient QSAMPLE preparation scheme using the aforementioned tools.

***Contributions*:** We present a new end-to-end framework for preparing the QSAMPLE, replacing the traditional QPE-based reflection procedure with a selective phase quantum circuit, built from a generalized quantum signal processing (GQSP)-based method using only one ancilla qubit in the working register [16], [17]. Next, we construct the selective

phase compilers and bound the operator norm of their deviation from the ideal selective phase operator, and then rigorously bound the trace distance when fed into each stage of the FPAA subroutine. We then assemble our new end-to-end quantum algorithm, which produces a QSAMPLE within any prescribed trace distance using

$$\widetilde{\mathcal{O}}\left(\frac{\ell}{\sqrt{p_{\min}}\Delta_{\min}}\right) \tag{2}$$

queries to the oracle $O_{\mathcal{P}}$ and analyse the qubit complexity. We then conduct a comprehensive resource-analysis simulation showing how our query and qubit complexity evolve with the trace distance, compared with [4]. Finally, we apply our framework to prepare a QSAMPLE of the Gibbs state, and by combining it with the schedule-length bound from [24], we obtain an ancilla efficient preparation scheme for the Gibbs QSAMPLE.

II. PRELIMINARIES

Let $\mathcal{P}$ be a reversible ergodic Markov chain on Ω, with transition matrix P, stationary distribution π, and spectral gap $\delta(\mathcal{P}) := 1-\lambda_2(P)$. Here $\lambda_2(P)$ is the second-largest algebraic eigenvalue, not the second-largest eigenvalue in magnitude; $\delta(\mathcal{P})$ is therefore the algebraic (Poincaré) gap. Define the step isometry

$$\mathcal{T}\left|x\right\rangle := \left|x\right\rangle \sum_{y\in\Omega} \sqrt{P(x,y)}\left|y\right\rangle, \qquad \mathcal{T}^\dagger\mathcal{T} = \mathbb{I}, \tag{3}$$

and the swap $S\left|x,y\right\rangle = \left|y,x\right\rangle$.

Definition II.1 (Qubitized Szegedy walk)**.** *The qubitized Szegedy walk associated with $\mathcal{P}$ is*

$$\mathcal{W} := (2\mathcal{T}\mathcal{T}^\dagger - \mathbb{I})S. \tag{4}$$

The stationary state lifts to

$$\left|\psi\right\rangle := \mathcal{T}\left|\pi\right\rangle = \sum_{x,y\in\Omega} \sqrt{\pi(x)P(x,y)}\ \left|x,y\right\rangle \tag{5}$$

in $\mathcal{H}_{\mathcal{W}} \cong \mathbb{C}^{|\Omega|}\otimes\mathbb{C}^{|\Omega|}$. We assume a coherently controllable unitary oracle satisfying

$$O_{\mathcal{P}} : \left|x\right\rangle\left|0\right\rangle \mapsto \sum_{y\in\Omega} \sqrt{P(x,y)}\left|x\right\rangle\left|y\right\rangle. \tag{6}$$

Let a be the one-qubit GQSP signal register, A the system/QSAMPLE register, and B the transition register. We place all ideal and compiled operations on one common space using the clean embedding

$$J\left|v\right\rangle := \left|0\right\rangle_a\left|v\right\rangle_A\left|0\right\rangle_B. \tag{7}$$

One transition-oracle query means one forward or inverse use of $O_{\mathcal{P}}$, controlled when required. Thus

$$\mathcal{W} = \big(2\mathcal{T}\mathcal{T}^\dagger - \mathbb{I}\big)S = \Big[\, O_{\mathcal{P}}\big(\mathbb{I}\otimes(2\left|0\right\rangle\left\langle 0\right| - \mathbb{I})\big)O_{\mathcal{P}}^\dagger\,\Big]S, \tag{8}$$

so a controlled walk uses one forward and one inverse oracle query, up to controlled swaps and elementary gates.

Definition II.2 (Phase gap)**.** *Let $\mathcal{W}$ be the qubitized Szegedy walk associated with $\mathcal{P}$. Its phase gap is*

$$\Delta_{\mathcal{W}} := \min\{|\theta| : \theta\neq 0,\ e^{i\theta}\in\operatorname{spec}(\mathcal{W}),\ \theta\in(-\pi,\pi]\}. \tag{9}$$

Szegedy spectral mapping gives

$$\begin{aligned}\Delta_{\mathcal{W}} &= \arccos(\lambda_2(P))\\ &\in \Omega\left(\sqrt{\delta(\mathcal{P})}\right).\end{aligned} \tag{10}$$

Let Π_0 denote the orthogonal projector onto the 1-eigenspace of $\mathcal{W}$, i.e. $\operatorname{Im}(\Pi_0) = \ker(\mathcal{W}-\mathbb{I})$. The full 1-eigenspace can contain vectors outside the active Szegedy subspace, so Π_0 need not be rank one. What the algorithm needs is the following restricted identity.

Lemma II.3 (Stationary-space intertwining)**.** *For a reversible ergodic chain $\mathcal{P}$, with $\Pi_\pi := \left|\pi\right\rangle\left\langle\pi\right|$,*

$$\Pi_0\mathcal{T} = \mathcal{T}\Pi_\pi. \tag{11}$$

Proof. Let $\mathcal{K} := \operatorname{Im}(\mathcal{T}) + S\operatorname{Im}(\mathcal{T})$. Directly from $\mathcal{W} = (2\mathcal{T}\mathcal{T}^\dagger-\mathbb{I})S$, one has $\mathcal{W}(\mathcal{K})\subseteq\mathcal{K}$. Since $\mathcal{W}$ is unitary on a finite-dimensional space, $\mathcal{W}(\mathcal{K}) = \mathcal{K}$; consequently $\mathcal{K}$ is reducing and $\mathcal{W}(\mathcal{K}^\perp) = \mathcal{K}^\perp$. The discriminant $\mathcal{T}^\dagger S\mathcal{T}$ is similar to P, and ergodicity makes its 1-eigenvector $\left|\pi\right\rangle$ unique. Szegedy spectral mapping therefore makes $\left|\psi\right\rangle = \mathcal{T}\left|\pi\right\rangle$ the only 1-eigenvector of $\mathcal{W}$ in $\mathcal{K}$. Any additional 1-eigenvectors lie in $\mathcal{K}^\perp$, hence are orthogonal to $\operatorname{Im}(\mathcal{T})$. Projecting an arbitrary $\mathcal{T}\left|v\right\rangle$ onto the full 1-eigenspace gives $\left|\psi\right\rangle\left\langle\psi\,|\,\mathcal{T}v\right\rangle = \mathcal{T}\left|\pi\right\rangle\left\langle\pi\,|\,v\right\rangle$, proving (11). □

Consequently, the relevant spectral decomposition is

$$\mathcal{W} = \Pi_0 + \sum_{j\geq 1} e^{i\theta_j}\Pi_j, \qquad |\theta_j|\geq\Delta_{\mathcal{W}},$$

and Lemma II.3 implies $\mathcal{S}_\phi(\Pi_0)\mathcal{T} = \mathcal{T}\mathcal{S}_\phi(\Pi_\pi)$. For density operators ρ,σ on the same Hilbert space, define

$$D_{\mathrm{tr}}(\rho,\sigma) := \frac{1}{2}\|\rho-\sigma\|_1, \tag{12}$$

where $\|\cdot\|_1$ is the trace norm. For normalized pure states, this specializes to

$$D_{\mathrm{tr}}(|\alpha\rangle\langle\alpha|, |\beta\rangle\langle\beta|) = \sqrt{1-|\left\langle\alpha\,|\,\beta\right\rangle|^2}, \tag{13}$$

and $D_{\mathrm{tr}}(|\alpha\rangle, |\beta\rangle)$ denotes this trace distance between the corresponding rank-one density operators. The total variation distance (TVD) of probability distributions α,β on Ω is

$$d_{\mathrm{TV}}(\alpha,\beta) := \frac{1}{2}\sum_{x\in\Omega}|\alpha(x)-\beta(x)|. \tag{14}$$

It can be shown that $d_{\mathrm{TV}}(\alpha,\beta)\leq D_{\mathrm{tr}}(|\alpha\rangle, |\beta\rangle)$. Throughout we also denote $\|\cdot\|$ as the operator norm.

Promised inputs and space convention. For every schedule stage we assume certified lower bounds on the adjacent squared overlap and walk phase gap. The cost of obtaining these promises is not included; proving a useful chain-gap bound can dominate an application, and a conservative bound directly increases the filter degree. "One ancilla" below means one

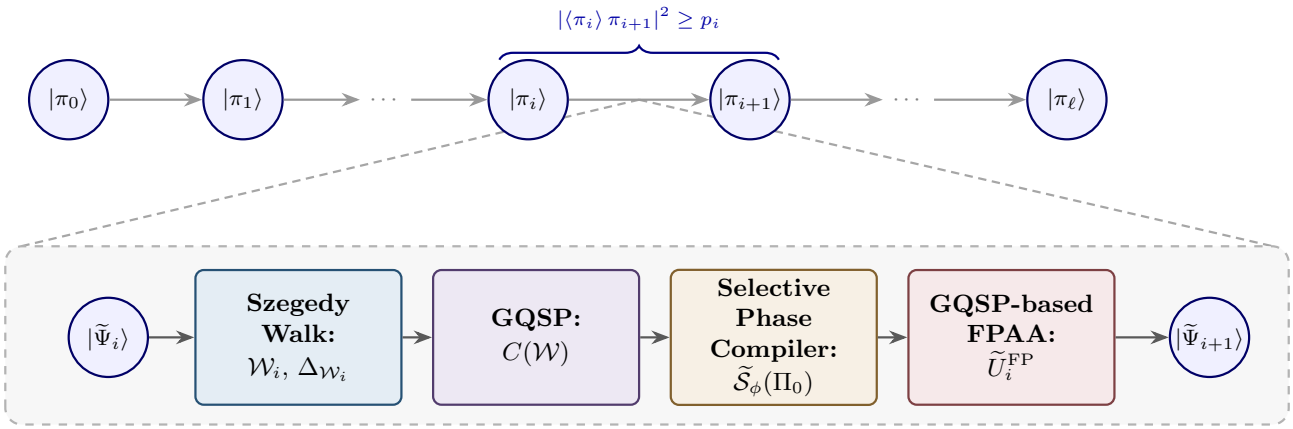


Fig. 1. Architecture of the constant-ancilla QSAMPLE preparation algorithm. **Top row:** the cooling schedule advances through ℓ stages of stationary states $|\pi_0\rangle \to \cdots \to |\pi_\ell\rangle$. **Bottom row:** each transition $|\widetilde{\pi}_i\rangle \to |\widetilde{\pi}_{i+1}\rangle$ uses the qubitized Szegedy walk, to build a GQSP-based selective-phase compiler, to then use FPAA subroutine to transport the state.

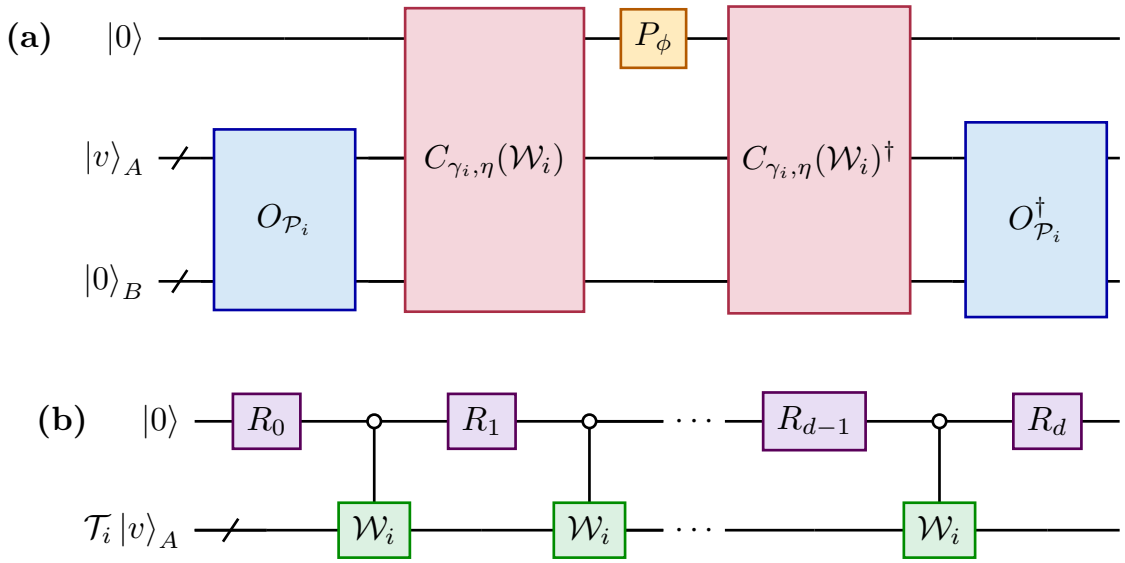


Fig. 2. Complete common-interface selective-phase compiler. (a) The transition oracle lifts $J|v\rangle$ into walk space, the GQSP block computes the zero-eigenphase filter, P_ϕ phases its accepted branch, and both are uncomputed. (b) The one-signal-qubit implementation of $C_{\gamma_i,\eta}(\mathcal{W}_i)$ acts on the lifted state $\mathcal{T}_i|v\rangle_A = O_{\mathcal{P}_i}(|v\rangle_A|0\rangle_B)$ arising in (a) and uses d zero-controlled walk calls. Open control circles denote control on the signal state $|0\rangle$. On the clean column, $\|\widetilde{\mathcal{S}}_{i,\phi}^{(\eta)}J - J\mathcal{S}_\phi(\Pi_{\pi_i})\| \leq |e^{i\phi}-1|\eta$.

clean GQSP signal qubit beyond the two Ω-registers. Any workspace internal to controlled-$O_{\mathcal{P}}$, arithmetic, or classical phase synthesis is excluded and must be added to the total implementation width.

III. Selective Phase Compiler

FPAA needs adjustable phases about both its source and target states, rather than only a π-reflection. For an orthogonal projector Π, define

Definition III.1 (Selective phase)**.** *For $\phi \in \mathbb{R}$, the selective phase about* $\mathrm{Im}(\Pi)$ *is*

$$\mathcal{S}_\phi(\Pi) := \mathbb{I} + (e^{i\phi} - 1)\Pi. \tag{15}$$

It applies $e^{i\phi}$ on the selected subspace and the identity on its complement; $\phi = \pi$ gives a reflection.

The construction has three transparent steps. First, a one-qubit GQSP circuit computes a block $\Upsilon(\mathcal{W})$ that is the identity on eigenphase zero and is suppressed on phases separated by the certified lower bound $0 < \gamma \leq \Delta_{\mathcal{W}}$. Second, a phase P_ϕ is applied only to the "accepted" signal-qubit branch and the filter is uncomputed. Finally, Lemma II.3 transports the operation from the lifted walk space to $|\pi\rangle$. Concretely,

$$\widetilde{\mathcal{S}}_\phi^{\mathrm{walk}}(\Pi_0) := C_{\gamma,\varepsilon}(\mathcal{W})^\dagger(P_\phi \otimes \mathbb{I})C_{\gamma,\varepsilon}(\mathcal{W}). \tag{16}$$

Here $P_\phi = e^{i\phi}|0\rangle\langle 0| + |1\rangle\langle 1|$.

Corollary III.2 (One-qubit GQSP filter)**.** *For any certified $0 < \gamma \le \Delta_{\mathcal{W}}$ and $\varepsilon^{\mathcal{W}} \in (0,1)$, there are degree-$d$ polynomials Υ, Φ and a one-signal-qubit unitary $C_{\gamma,\varepsilon}(\mathcal{W})$ such that*

$$C_{\gamma,\varepsilon}(\mathcal{W})(|0\rangle \otimes \mathbb{I}) = |0\rangle \otimes \Upsilon(\mathcal{W}) + |1\rangle \otimes \Phi(\mathcal{W}), \tag{17}$$

$$\|\Upsilon(\mathcal{W}) - \Pi_0\| \le \varepsilon^{\mathcal{W}},$$

$$\Upsilon(\mathcal{W})^\dagger \Upsilon(\mathcal{W}) + \Phi(\mathcal{W})^\dagger \Phi(\mathcal{W}) = \mathbb{I}, \tag{18}$$

One explicit choice is

$$t_\gamma := \left\lceil \frac{\mathrm{e}}{\sin(\gamma/2)} \right\rceil, \qquad n_\varepsilon := \left\lceil \ln \frac{1}{\varepsilon^{\mathcal{W}}} \right\rceil,$$
$$\Upsilon_{\gamma,\varepsilon}(z) := \left(\frac{1}{t_\gamma} \sum_{k=0}^{t_\gamma - 1} z^k \right)^{n_\varepsilon}. \tag{19}$$

Its degree and the exact number of controlled-$\mathcal{W}$ calls in $C_{\gamma,\varepsilon}$ are

$$d = (t_\gamma - 1) n_\varepsilon = \mathcal{O}\left(\frac{1}{\gamma} \log \frac{2}{\varepsilon^{\mathcal{W}}} \right). \tag{20}$$

Proof. The averaged-power family follows the geometric-series filter of Ref. [17]; the constant in t_γ is chosen directly from the bound below. The filter has $\Upsilon_{\gamma,\varepsilon}(1) = 1$, and the triangle inequality gives $|\Upsilon_{\gamma,\varepsilon}(z)| \le 1$ for $|z| = 1$. For $\gamma \le |\theta| \le \pi$,

$$\left| \frac{1}{t_\gamma} \sum_{k=0}^{t_\gamma - 1} e^{ik\theta} \right| = \frac{|\sin(t_\gamma \theta/2)|}{t_\gamma |\sin(\theta/2)|} \le \frac{1}{t_\gamma \sin(\gamma/2)} \le \mathrm{e}^{-1}. \tag{21}$$

Therefore $|\Upsilon_{\gamma,\varepsilon}(e^{i\theta})| \le \mathrm{e}^{-n_\varepsilon} \le \varepsilon^{\mathcal{W}}$. Functional calculus gives

$$\Upsilon(\mathcal{W}) - \Pi_0 = \sum_{j \ge 1} \Upsilon(e^{i\theta_j}) \Pi_j,$$

whose norm is at most $\varepsilon^{\mathcal{W}}$. Fejér–Riesz factorization of the nonnegative trigonometric polynomial $1 - |\Upsilon|^2$ supplies a degree-at-most-d complement Φ; the GQSP theorem then realizes (17)–(18) with exactly d controlled-$\mathcal{W}$ calls and one signal qubit [16]. □

We use the zero-controlled signal convention: each controlled $\mathcal{W}$ acts when the signal is $|0\rangle$, interleaved with single-qubit rotations $R_0, \dots, R_d \in \mathrm{SU}(2)$. The GQSP synthesis theorem supplies these rotations for Υ; no additional phase-estimation register is introduced.

Proposition III.3 (Compiled selective phase)**.** *The circuit* (16) *satisfies*

$$\left\| \widetilde{\mathcal{S}}_\phi^{\mathrm{walk}}(\Pi_0)(|0\rangle \otimes \mathbb{I})\mathcal{T} - (|0\rangle \otimes \mathbb{I})\mathcal{T}\mathcal{S}_\phi(\Pi_\pi) \right\| \le 2|\sin(\phi/2)|\varepsilon^{\mathcal{W}} \le 2\varepsilon^{\mathcal{W}}. \tag{22}$$

It uses one signal qubit, exactly $2d$ controlled-$\mathcal{W}$ or controlled-$\mathcal{W}^\dagger$ queries, and $4d$ transition-oracle queries. Here d is (20)*; each walk query contains one forward and one inverse transition-oracle call.*

Proof. Suppress tensor-product identities and set $\mathcal{Q} := \Upsilon(\mathcal{W})^\dagger \Upsilon(\mathcal{W})$. Equation (17) makes the signal-$|0\rangle$ block after compute–phase–uncompute

$$\begin{aligned} \mathcal{A} &= \langle 0| \widetilde{\mathcal{S}}_\phi^{\mathrm{walk}}(\Pi_0) |0\rangle \\ &= e^{i\phi} \Upsilon(\mathcal{W})^\dagger \Upsilon(\mathcal{W}) + \Phi(\mathcal{W})^\dagger \Phi(\mathcal{W}) \\ &= \mathbb{I} + (e^{i\phi} - 1)\mathcal{Q}. \end{aligned} \tag{23}$$

Thus the wrapper replaces the ideal projector by the positive operator

$$\mathcal{Q} = \Pi_0 + \sum_{j \ge 1} |\Upsilon(e^{i\theta_j})|^2 \Pi_j,$$

so $0 \le \mathcal{Q} - \Pi_0 \le (\varepsilon^{\mathcal{W}})^2 (\mathbb{I} - \Pi_0)$. Let $\mathcal{B} := \langle 1| \widetilde{\mathcal{S}}_\phi^{\mathrm{walk}}(\Pi_0) |0\rangle$. Unitarity of the first block column gives

$$\mathcal{B}^\dagger \mathcal{B} = \mathbb{I} - \mathcal{A}^\dagger \mathcal{A} = |e^{i\phi} - 1|^2 (\mathcal{Q} - \mathcal{Q}^2).$$

For

$$\mathcal{E}_\phi := \widetilde{\mathcal{S}}_\phi^{\mathrm{walk}}(\Pi_0)(|0\rangle \otimes \mathbb{I}) - (|0\rangle \otimes \mathbb{I})\mathcal{S}_\phi(\Pi_0).$$

the signal branches are orthogonal and

$$\mathcal{E}_\phi^\dagger \mathcal{E}_\phi = |e^{i\phi} - 1|^2 (\mathcal{Q} - \Pi_0).$$

Hence $\|\mathcal{E}_\phi\| \le |e^{i\phi} - 1| \varepsilon^{\mathcal{W}} = 2|\sin(\phi/2)| \varepsilon^{\mathcal{W}}$. Right multiplication by the isometry $\mathcal{T}$, followed by Lemma II.3, proves (22). □

Using the clean embedding J defined in Section II, place the ideal and implemented operations on the common joint space. For chain j and accuracy η, let $\Pi_{0,j}$ be the full 1-eigenspace projector of $\mathcal{W}_j$, and define

$$\widetilde{\mathcal{S}}_{j,\phi}^{(\eta)} := (\mathbb{I}_a \otimes O_{\mathcal{P}_j}^\dagger) \widetilde{\mathcal{S}}_\phi^{\mathrm{walk}}(\Pi_{0,j})(\mathbb{I}_a \otimes O_{\mathcal{P}_j}). \tag{24}$$

The middle circuit uses $C_{\gamma_j,\eta}(\mathcal{W}_j)$. Proposition III.3 is equivalently the restricted map bound

$$\left\| \widetilde{\mathcal{S}}_{j,\phi}^{(\eta)} J - J\mathcal{S}_\phi(\Pi_{\pi_j}) \right\| \le 2|\sin(\phi/2)|\eta. \tag{25}$$

For the corresponding degree-d filter, $\widetilde{\mathcal{S}}_{j,\phi}^{(\eta)}$ uses $2d$ controlled walk calls and $4d + 2$ transition-oracle calls; the final two queries are the lift and unlift in (24).

IV. Fixed-Point Amplitude Amplification

FPAA transports $|\pi_i\rangle$ to $|\pi_{i+1}\rangle$ using only a lower bound $p_i \le q_i$, avoiding over-rotation while retaining the quadratic $1/\sqrt{p_i}$ dependence [14]. In contrast, the $\pi/3$ fixed-point procedure analyzed in Ref. [4] scales as $1/p_i$.

Theorem IV.1 (FPAA [14], restated)**.** *For $q_i \ge p_i > 0$ and $\varepsilon_i^{\mathrm{FP}} \in (0,1)$, choose an odd $L_i \ge 3$ such that, with $\xi_i^{-1} := T_{1/L_i}(1/\varepsilon_i^{\mathrm{FP}})$,*

$$1 - \xi_i^2 \le p_i. \tag{26}$$

Here $T_\nu(x) = \cosh(\nu \operatorname{arcosh} x)$ for $x \ge 1$. The smallest such odd sequence parameter L_i obeys

$$L_i = \mathcal{O}\left(\frac{1}{\sqrt{p_i}} \log \frac{2}{\varepsilon_i^{\mathrm{FP}}} \right) \tag{27}$$

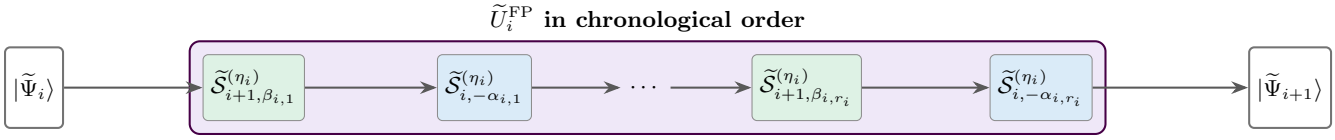


Fig. 3. One FPAA transport stage in the fixed common embedding. In chronological order, the Yoder–Low–Chuang FPAA unitary alternates a compiled target-chain phase $\widetilde{\mathcal{S}}^{(\eta_i)}_{i+1,\beta_{i,j}}$ with a compiled source-chain phase $\widetilde{\mathcal{S}}^{(\eta_i)}_{i,-\alpha_{i,j}}$. Every wrapper begins and ends in the same J-encoded space, so the phase-dependent clean-column bounds compose without a hidden change of register convention.

and its phases are explicit. Writing $r_i = (L_i - 1)/2$*, for* $1 \le j \le r_i$ *take*

$$\alpha_{i,j} = -\beta_{i,r_i-j+1} = 2\cot^{-1}\left[\tan\frac{2\pi j}{L_i}\sqrt{1-\xi_i^2}\right]. \tag{28}$$

Here $\cot^{-1}\colon \mathbb{R} \to (0,\pi)$ *denotes the principal arccotangent, and phases are understood modulo* 2π*. Set* $V_{i,2j-1} := \mathcal{S}_{\beta_{i,j}}(\Pi_{\pi_{i+1}})$ *and* $V_{i,2j} := \mathcal{S}_{-\alpha_{i,j}}(\Pi_{\pi_i})$*. Then, with* $m_i := L_i - 1$*,*

$$U_i^{\mathrm{FP}} := V_{i,m_i}\cdots V_{i,1} \tag{29}$$

satisfies

$$D_{\mathrm{tr}}\big(U_i^{\mathrm{FP}}\,|\pi_i\rangle\,,|\pi_{i+1}\rangle\big) \le \varepsilon_i^{\mathrm{FP}}.$$

Products act right-to-left: in every pair the target phase $V_{i,2j-1}$ *acts first and the source phase* $V_{i,2j}$ *second, matching the generalized Grover iterate of Ref. [14] up to its irrelevant global minus sign.*

Corollary IV.2 (Phase-weighted compiled FPAA)**.** *Choose* L_i *and its phases as in Theorem IV.1. Replace every* $V_{i,k}$ *by its lifted-back compiler* $\widetilde{V}_{i,k}$*. Let the source filter in pair* j *have accuracy* $\eta^{\mathrm{s}}_{i,j}$*, and the target filter have accuracy* $\eta^{\mathrm{t}}_{i,j}$*. Set* $\widetilde{U}_i^{\mathrm{FP}} := \widetilde{V}_{i,m_i}\cdots\widetilde{V}_{i,1}$*. Then*

$$D_{\mathrm{tr}}\left(\widetilde{U}_i^{\mathrm{FP}}J\,|\pi_i\rangle\,,J\,|\pi_{i+1}\rangle\right) \le \varepsilon_i^{\mathrm{FP}} + 2\sum_{j=1}^{r_i}\left|\sin\frac{\alpha_{i,j}}{2}\right|\eta^{\mathrm{s}}_{i,j} + 2\sum_{j=1}^{r_i}\left|\sin\frac{\beta_{i,j}}{2}\right|\eta^{\mathrm{t}}_{i,j}. \tag{30}$$

Proof. The clean-column estimate is not a global operator bound, so we hybridize through ideal clean prefixes. Let $\widetilde{V}_{>k} := \widetilde{V}_{i,m_i}\cdots\widetilde{V}_{i,k+1}$ and $V_{<k} := V_{i,k-1}\cdots V_{i,1}$, with empty products equal to the identity. Then the exact identity

$$\widetilde{U}_i^{\mathrm{FP}}J - JU_i^{\mathrm{FP}} = \sum_{k=1}^{m_i}\widetilde{V}_{>k}(\widetilde{V}_{i,k}J - JV_{i,k})V_{<k}$$

is well typed on the common joint space. All outer factors are unitary, and (25) applies because the right prefix is ideal and therefore remains in the clean image of J. Applying its phase-dependent bound separately to every source and target phase gives the sum in (30). Trace distance between the corresponding pure joint states is no larger than their vector distance. The triangle inequality and Theorem IV.1 complete the proof. □

For later use, define the computable phase weight

$$A_i := \sum_{j=1}^{r_i}\left(\left|\sin\frac{\alpha_{i,j}}{2}\right| + \left|\sin\frac{\beta_{i,j}}{2}\right|\right) \le L_i - 1. \tag{31}$$

Indeed, with $s_i := \sqrt{1-\xi_i^2} \le 1$, phase-list symmetry and the $j = r_i$ term give $A_i \ge 2[1 + s_i^2\tan^2(\pi/L_i)]^{-1/2} \ge 2\cos(\pi/L_i) \ge 1$; hence $\eta_i \in (0,1)$. If all filters at stage i use accuracy η_i, Corollary IV.2 becomes $D_{\mathrm{tr}} \le \varepsilon_i^{\mathrm{FP}} + 2A_i\eta_i$.

V. End-to-End QSample Preparation Framework

At stage i, FPAA alternates lifted-back compilers for chains i and $i+1$; it is not a single pass through two reflections. The same signal qubit and transition register are reused throughout. Algorithm 1 exposes the required classical promises and the error allocation.

Algorithm 1: QSample Preparation

Input: Controlled circuits $\{O_{\mathcal{P}_i}, O_{\mathcal{P}_i}^\dagger\}_{i=0}^{\ell}$; certified $p_i > 0$ with $|\langle\pi_i\,|\,\pi_{i+1}\rangle|^2 \ge p_i$ for $i < \ell$ and $\Delta_{\mathcal{W}_j} \ge \gamma_j > 0$ for $j \le \ell$; $|\pi_0\rangle$; error $\varepsilon \in (0,1)$.

Output: Joint state $|\widetilde{\Psi}_\ell\rangle$ satisfying $D_{\mathrm{tr}}(|\widetilde{\Psi}_\ell\rangle\,, J\,|\pi_\ell\rangle) \le \varepsilon$.

Initialize $|\widetilde{\Psi}_0\rangle \leftarrow J\,|\pi_0\rangle$;
for $i \leftarrow 0$ **to** $\ell - 1$ **do**
  Set $\varepsilon_i = \varepsilon/\ell$ and $\varepsilon_i^{\mathrm{FP}} = \varepsilon_i/2$, then choose the smallest odd $L_i \ge 3$ satisfying (26) and compute its phases;
  Compute A_i from (31) and set $\eta_i = \varepsilon_i/(4A_i)$;
  Synthesize the i and $i+1$ filters using γ_i, γ_{i+1} and accuracy η_i;
  $|\widetilde{\Psi}_{i+1}\rangle \leftarrow \widetilde{U}_i^{\mathrm{FP}}\,|\widetilde{\Psi}_i\rangle$;
return $|\widetilde{\Psi}_\ell\rangle$;

For the explicit filter (19), define the stage-i, chain-j degree

$$d_{i,j} := \left(\left\lceil\frac{\mathrm{e}}{\sin(\gamma_j/2)}\right\rceil - 1\right)\left\lceil\ln\frac{1}{\eta_i}\right\rceil, \qquad j \in \{i, i+1\}. \tag{32}$$

Counting each forward or inverse use of $O_{\mathcal{P}_j}$, controlled when required, as one transition-oracle query, stage i uses exactly

$$Q_i = 2(L_i - 1)\big(d_{i,i} + d_{i,i+1} + 1\big) \tag{33}$$

queries, including every lift and unlift.

Theorem V.1 (End-to-end preparation)**.** *Let $\ell \geq 1$, let $\varepsilon \in (0,1)$, and let $\{\mathcal{P}_i\}_{i=0}^{\ell}$ be reversible ergodic chains on Ω, with stationary* QSAMPLES *$|\pi_i\rangle$. Suppose that, for $0 \leq i < \ell$,*

$$q_i := |\langle \pi_i | \pi_{i+1} \rangle|^2 \geq p_i > 0, \tag{34}$$

and that $\Delta_{\mathcal{W}_i} \geq \gamma_i > 0$ for $0 \leq i \leq \ell$, where p_i, γ_i are supplied certified bounds. Define

$$\gamma_i^\star := \min\{\gamma_i, \gamma_{i+1}\}, \quad p_{\min} := \min_{i<\ell} p_i, \quad \gamma_{\min} := \min_{i\leq\ell} \gamma_i.$$

Given exact preparation of $|\pi_0\rangle$ and coherently controllable circuits $O_{\mathcal{P}_i}, O_{\mathcal{P}_i}^\dagger$, Algorithm 1 produces a joint pure state $|\widetilde{\Psi}_\ell\rangle$ such that

$$D_{\mathrm{tr}}\Big(|\widetilde{\Psi}_\ell\rangle, J|\pi_\ell\rangle\Big) \leq \varepsilon. \tag{35}$$

Its transition-oracle query count, with control when required, is

$$\mathcal{O}\left(\sum_{i=0}^{\ell-1} \frac{\log(4\ell/\varepsilon)}{\sqrt{p_i}\,\gamma_i^\star} \log\left(\mathrm{e} + \frac{\ell}{\varepsilon\sqrt{p_i}} \log\frac{4\ell}{\varepsilon}\right)\right). \tag{36}$$

In particular, this is

$$\widetilde{\mathcal{O}}\left(\frac{\ell}{\sqrt{p_{\min}}\,\gamma_{\min}}\right).$$

If $\widetilde{\rho}_\ell := \mathrm{Tr}_{aB} |\widetilde{\Psi}_\ell\rangle\langle\widetilde{\Psi}_\ell|$, then

$$D_{\mathrm{tr}}(\widetilde{\rho}_\ell, |\pi_\ell\rangle\langle\pi_\ell|) \leq \varepsilon, \qquad d_{\mathrm{TV}}(\widetilde{\pi}_\ell, \pi_\ell) \leq \varepsilon,$$

where $\widetilde{\pi}_\ell$ is the outcome distribution obtained by measuring A. Outside oracle-internal scratch space, the peak width is $2\lceil\log_2|\Omega|\rceil + 1$ qubits.

Proof. Set $\varepsilon_i = \varepsilon/\ell$, $\varepsilon_i^{\mathrm{FP}} = \varepsilon_i/2$, and $\eta_i = \varepsilon_i/(4A_i)$. Corollary IV.2 gives

$$D_{\mathrm{tr}}\Big(\widetilde{U}_i^{\mathrm{FP}} J|\pi_i\rangle, J|\pi_{i+1}\rangle\Big) \leq \varepsilon_i.$$

Theorem IV.1, with $\varepsilon_i^{\mathrm{FP}} = \varepsilon/(2\ell)$, gives

$$L_i = \mathcal{O}\left(p_i^{-1/2} \log\frac{4\ell}{\varepsilon}\right).$$

Each compiled phase uses the i- or $i+1$-chain filter and therefore

$$\mathcal{O}\left(\frac{1}{\gamma_i^\star}\log\frac{1}{\eta_i}\right)$$

controlled walks. Since $\eta_i = \varepsilon/(4\ell A_i)$ and $A_i \leq L_i - 1$, $\log(1/\eta_i) = \log(4\ell A_i/\varepsilon)$ is

$$\mathcal{O}\left(\log\left(\mathrm{e} + \frac{\ell}{\varepsilon\sqrt{p_i}}\log\frac{4\ell}{\varepsilon}\right)\right).$$

Multiplying by $L_i - 1$ and summing the stages yields (36). Equation (33) gives the corresponding constant-resolved count for the explicit filter.

For the actual recurrence $|\widetilde{\Psi}_{i+1}\rangle := \widetilde{U}_i^{\mathrm{FP}}|\widetilde{\Psi}_i\rangle$, unitary invariance and the triangle inequality give

$$D_{\mathrm{tr}}(|\widetilde{\Psi}_{i+1}\rangle, J|\pi_{i+1}\rangle) \leq D_{\mathrm{tr}}(|\widetilde{\Psi}_i\rangle, J|\pi_i\rangle) + \varepsilon_i.$$

The initial term is zero, proving (35) by induction. Contractivity under partial trace gives the system-register statement, and contractivity under measurement gives the TVD statement. The walk uses two Ω-registers and the compiler reuses one signal qubit; any workspace inside the supplied oracle circuits is excluded. □

TABLE I
FIXED-SCHEDULE COMPARISON WITH THE COHERENT-QPE/$\pi/3$ CONSTRUCTION OF REF. [4].

	Ref. [4]	**This work**
Queries	$\widetilde{\mathcal{O}}\Big(\frac{\ell}{p_{\min}\gamma_{\min}}\Big)$	$\widetilde{\mathcal{O}}\Big(\frac{\ell}{\sqrt{p_{\min}}\,\gamma_{\min}}\Big)$
Szegedy-walk qubits	$2\lceil\log_2|\Omega|\rceil$	$2\lceil\log_2|\Omega|\rceil$
Ancilla overhead	$\mathcal{O}\Big(\log\frac{1}{\gamma_{\min}}\log\frac{\ell}{\varepsilon p_{\min}}\Big)$	1
Oracle-internal scratch	excluded	excluded

Here one query is one forward or inverse use of $O_{\mathcal{P}_i}$, controlled when required, and both columns exclude oracle-internal scratch space. The comparison holds the prescribed schedule and access model fixed: schedule refinement may exchange a larger ℓ for better adjacent overlaps. Single-ancilla iterative QPE and coherent low-ancilla phase-estimation alternatives exist [21], [22]; Table I concerns the particular coherent-QPE reflection analyzed in Ref. [4].

A. A regime with subconstant adjacent overlap

The improvement over the Ref. [4] $1/p_i$ bound is asymptotic outside the well-balanced schedule used in Section VI. Consider $\Omega = \{0,1\}^n$ and $H(x) = \sum_{k=1}^n x_k$. Its Gibbs distribution is $\mu_\beta = \mathrm{Ber}(r_\beta)^{\otimes n}$, where $r_\beta = (1+\mathrm{e}^\beta)^{-1}$. Between $\beta = 0$ and $\beta > 0$, the squared QSAMPLE overlap is

$$q(\beta) = \left[\frac{1 + \mathrm{sech}(\beta/2)}{2}\right]^n. \tag{37}$$

For $n \geq 2$, choose

$$\beta_n = 2\,\mathrm{arcosh}\left(\frac{1}{2n^{-1/n} - 1}\right) = 4\sqrt{\frac{\ln n}{n}}\left(1 + \mathcal{O}\left(\frac{\ln n}{n}\right)\right). \tag{38}$$

Then $q(\beta_n) = 1/n$ exactly; take $p_0 = q(\beta_n)$.

Let $\mathcal{P}_\beta$ be the lazified random-scan heat-bath chain with transition matrix $P_\beta = (I + G_\beta)/2$. In the μ_β-orthogonal product basis, a mode supported on $S \subseteq [n]$ has eigenvalue $1 - |S|/(2n)$; hence

$$\delta(\mathcal{P}_\beta) = \frac{1}{2n}, \qquad \Delta_{\mathcal{W}_\beta} = \arccos\left(1 - \frac{1}{2n}\right) = \Theta(n^{-1/2}) \tag{39}$$

at both endpoints. On this one-stage schedule, the upper bound of Ref. [4] is $\widetilde{\mathcal{O}}(n^{3/2})$, whereas Theorem V.1 gives $\widetilde{\mathcal{O}}(n)$. This product model transparently separates the Ref. [4] upper bound from the present GQSP/FPAA upper bound. The transition oracle is not a target-state oracle: from x, a row is supported only on x and its n one-bit neighbors, whereas μ_{β_n} has full support. Nevertheless, this is not a query lower bound or a claim of sampling hardness: its target QSAMPLE can independently be prepared by single-qubit rotations.

B. Full-state small-instance validation

We use the open 2×4 ladder with ten edges,

$$H(x) = \sum_{(u,v)\in E} \mathbf{1}[x_u \neq x_v],$$

Configurations use row-major lexicographic bit ordering. The inverse-temperature schedule is $0, 0.3, 0.6, 0.9, 1.2$. This disagreement-count normalization is equivalent, up to an additive constant, to the conventional $-J\sum_{(u,v)} s_u s_v$ Hamiltonian with $J = 1/2$. Random-scan heat-bath dynamics chooses one of eight sites uniformly and resamples it from its exact conditional Gibbs law. Its lazified chain is $\mathcal{P}_\beta$, with transition matrix $P_\beta = (I + G_\beta)/2$.

All 2^8 configurations are enumerated. To fix the transition oracle away from its promised clean column, set $|p_x\rangle := \sum_y \sqrt{P(x,y)}\,|y\rangle$ and use the deterministic row-wise Householder extension

$$\begin{aligned} O_{\mathcal{P}} &= \sum_x |x\rangle\langle x| \otimes H_x, \\ H_x &= I - 2\,|u_x\rangle\langle u_x|\,, \\ |u_x\rangle &= \frac{|0\rangle - |p_x\rangle}{\| \,|0\rangle - |p_x\rangle\,\|}. \end{aligned} \tag{40}$$

with $H_x = I$ when the denominator vanishes. Across the five chains, the maximum row-sum, stationarity, detailed-balance, and projector-intertwining operator-norm residuals are respectively 4.44×10^{-16}, 1.39×10^{-17}, 1.08×10^{-19}, and 1.70×10^{-14}.

For $\varepsilon = 10^{-3}$, the phase-weighted allocation gives $L_i = 5$, $A_i = 2.28249$, $\varepsilon_i^{\mathrm{FP}} = 1.25 \times 10^{-4}$, and $\eta_i = 2.73824 \times 10^{-5}$ at every stage. For table compactness, write $\delta_i := \delta(\mathcal{P}_{\beta_i})$ and $\Delta_i := \Delta_{\mathcal{W}_i}$. Table II reports the resulting degrees, stagewise errors, and every controlled-walk, outer lift/unlift, and total transition-oracle call. Every table and figure distance and leakage value follows the approximate compiled phases $\widetilde{\mathcal{S}}_{i,\phi}^{(\eta_i)}$; ideal projectors are used only for sequence design and an audit control.

With $|\overline{\Psi}_{i+1}\rangle := |\widetilde{\Psi}_{i+1}\rangle / \|\widetilde{\Psi}_{i+1}\|_2$, the leakage probability is $\lambda_i^{\mathrm{leak}} := \sum_{a,x,y:\,(a,y)\neq(0,0)} |\langle a,x,y|\overline{\Psi}_{i+1}\rangle|^2$, which is evaluated as a sum of nonnegative terms rather than by subtraction. Binary64 values at or below $10\epsilon_{\mathrm{mach}}$, with $\epsilon_{\mathrm{mach}} = 2^{-52}$, are shown as $< 2.3 \times 10^{-15}$.

We then propagate the complete state in $\mathbb{C}^2 \otimes \mathbb{C}^{256} \otimes \mathbb{C}^{256}$ through every lift, filter, conditional phase, unfilter, unlift, and FPAA stage. Walk functions are applied from the discriminant spectrum, without forming the dense 65536×65536 walk. To make off-clean signal dynamics deterministic, the reference fixes, at each eigenvalue z, the unitary scalar completion

$$C(z) = \begin{pmatrix} \Upsilon(z) & -\sqrt{1-|\Upsilon(z)|^2} \\ \sqrt{1-|\Upsilon(z)|^2} & \overline{\Upsilon(z)} \end{pmatrix}.$$

This is a full-state spectral reference for the intended polynomial block, not an angle-level GQSP circuit simulation; the theorem's error envelope is independent of this completion.

At $\varepsilon = 10^{-3}$, the final joint and reduced trace distances are both 1.082×10^{-4}, the measured-system TVD is 1.889×10^{-5}, the clean-subspace leakage probability is below the 2.3×10^{-15} reporting threshold, and the propagated-state norm residual is 2.29×10^{-14}. The ideal-projector control gives $1.0816783746 \times 10^{-4}$, while the compiled-to-ideal full-state distance is 1.0847510×10^{-8}. The `qsample-validation` code, data, and scripts were used to generate the reported data. The routines use double-precision diagonalization rather than interval arithmetic; finite-precision GQSP rotation-synthesis error is excluded. The actual-phase hybrid bounds total 7.6122×10^{-7} versus a 5×10^{-4} allocation; this is one exact-promise instance, not an empirical scaling test.

VI. Application: Classical Gibbs QSample

For a classical Hamiltonian $H : \Omega_\mu \to \mathbb{R}$, let

$$\mu_\beta(x) := \frac{e^{-\beta H(x)}}{\mathcal{Z}(\beta)}, \qquad \mathcal{Z}(\beta) := \sum_x e^{-\beta H(x)}, \tag{41}$$

and $|\mu_\beta\rangle := \sum_x \sqrt{\mu_\beta(x)}\,|x\rangle$.

Corollary VI.1 (Classical Gibbs QSample)**.** *Let $n \geq 2$, let $H : \Omega_\mu \to \{0, \ldots, n\}$, fix a target inverse temperature $\beta_\star$, and suppose we are supplied a schedule*

$$0 = \beta_0 < \cdots < \beta_\ell = \beta_\star, \qquad \ell = \mathcal{O}\left(\sqrt{\ln|\Omega_\mu|\,\ln n}\right),$$

obeying the well-balanced guarantee of Ref. [24]. For each i, assume a reversible chain $\mathcal{P}_{\beta_i}$, with transition matrix P_{β_i} and stationary distribution μ_{β_i}, a coherently controllable transition circuit, and a certified walk phase-gap lower bound γ_i. Assume also exact preparation of the uniform initial state $|\mu_0\rangle = |\Omega_\mu|^{-1/2}\sum_{x\in\Omega_\mu}|x\rangle$. Then Algorithm 1 prepares a joint state whose reduced system state is within trace distance ε of $|\mu_{\beta_\star}\rangle$, using

$$\widetilde{\mathcal{O}}\left(\frac{\sqrt{\ln|\Omega_\mu|\,\ln n}}{\gamma_{\min}}\right) \tag{42}$$

transition-oracle queries, controlled when required, and one signal qubit beyond the two Ω_μ-registers, excluding oracle scratch.

Proof. The adjacent squared overlap is

$$|\,\langle \mu_{\beta_i} \,|\, \mu_{\beta_{i+1}} \rangle\,|^2 = \frac{\mathcal{Z}((\beta_i+\beta_{i+1})/2)^2}{\mathcal{Z}(\beta_i)\mathcal{Z}(\beta_{i+1})} \geq \frac{1}{15}$$

for the schedule of Ref. [24]. Thus $p_i = 1/15$ may be taken at every stage, so $p_{\min} = 1/15$, and (42) follows directly from Theorem V.1. Constructing the schedule and certifying the chain gaps are separate costs and are not included. □

If certified classical spectral-gap bounds satisfy $0 < \underline{\delta}_i \leq \delta(\mathcal{P}_{\beta_i})$ for $0 \leq i \leq \ell$, then $\lambda_2(P_{\beta_i}) \leq 1 - \underline{\delta}_i$, so one may choose $\gamma_i := \arccos(1 - \underline{\delta}_i) \leq \Delta_{\mathcal{W}_i}$. Since $1 - \cos\theta \leq \theta^2/2$, $1/\gamma_i \leq 1/\sqrt{2\underline{\delta}_i} = \mathcal{O}(1/\sqrt{\underline{\delta}_i})$. Writing $\underline{\delta}_{\min} := \min_{0\leq i\leq \ell} \underline{\delta}_i$, (42) therefore becomes $\widetilde{\mathcal{O}}(\sqrt{\ln|\Omega_\mu|\,\ln n}/\sqrt{\underline{\delta}_{\min}})$. Because $p_{\min}$ is constant here, replacing $1/p_{\min}$ by $1/\sqrt{p_{\min}}$ changes only a constant. The Gibbs corollary's asymptotic advantage

TABLE II

FULLY ENUMERATED OPEN 2×4 DISAGREEMENT-COUNT ISING BENCHMARK. PANEL (A) GIVES THE STAGE INPUTS. PANEL (B) FIXES $\varepsilon = 10^{-3}$, $\varepsilon_i^{\mathrm{FP}} = 1.25 \times 10^{-4}$, AND $A_i = 2.28249$, AND REPORTS FULL-STATE PROPAGATION UNDER THE APPROXIMATE COMPILED PHASES $\widetilde{\mathcal{S}}$. THE IDEAL PROJECTORS ARE USED ONLY FOR SEQUENCE DESIGN AND AN AUDIT CONTROL. N_{cW} COUNTS CONTROLLED WALK CALLS, N_{wrap} COUNTS OUTER $O_{\mathcal{P}_j}^{(\dagger)}$ CALLS, AND $Q_i = 2N_{\mathrm{cW}} + N_{\mathrm{wrap}}$. $\widetilde{D}_i^{\mathrm{local}}$ STARTS STAGE i FROM A FRESH $\mathcal{J}|\pi_i\rangle$; $\widetilde{D}_i^{\mathrm{cum}}$ COMPARES THE SEQUENTIAL PREFIX THROUGH STAGE i WITH $\mathcal{J}|\pi_{i+1}\rangle$.

(a) Stage inputs.

i	$\beta_i \to \beta_{i+1}$	$p_i = q_i$	(δ_i, δ_{i+1})	(Δ_i, Δ_{i+1})	L_i
0	$0.0 \to 0.3$	0.944415	$(0.062500, 0.039040)$	$(0.355421, 0.280346)$	5
1	$0.3 \to 0.6$	0.939133	$(0.039040, 0.021008)$	$(0.280346, 0.205340)$	5
2	$0.6 \to 0.9$	0.933454	$(0.021008, 0.009999)$	$(0.205340, 0.141532)$	5
3	$0.9 \to 1.2$	0.936311	$(0.009999, 0.004360)$	$(0.141532, 0.093413)$	5

(b) Compiled transport.

i	$(d_{i,i}, d_{i,i+1})$	η_i	$\widetilde{D}_i^{\mathrm{local}}$	$\widetilde{D}_i^{\mathrm{cum}}$	N_{cW}	N_{wrap}	Q_i	leakage
0	$(165, 209)$	2.74×10^{-5}	1.22×10^{-4}	1.22×10^{-4}	1496	8	3000	$< 2.3\times10^{-15}$
1	$(209, 286)$	2.74×10^{-5}	1.04×10^{-4}	8.18×10^{-5}	1980	8	3968	$< 2.3\times10^{-15}$
2	$(286, 418)$	2.74×10^{-5}	6.37×10^{-5}	6.41×10^{-5}	2816	8	5640	$< 2.3\times10^{-15}$
3	$(418, 638)$	2.74×10^{-5}	8.66×10^{-5}	1.08×10^{-4}	4224	8	8456	$< 2.3\times10^{-15}$

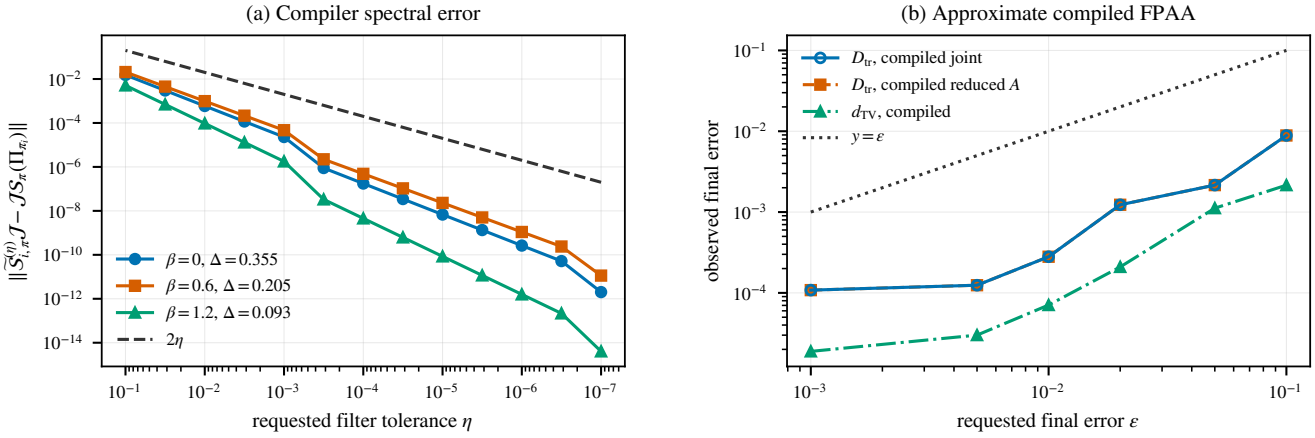


Fig. 4. Deterministic double-precision validation of the fixed-completion polynomial-block reference. (a) The restricted compiler error for $\phi = \pi$, evaluated on every nonstationary eigenphase of three benchmark walks, remains below the analytic 2η envelope. (b) Sequential full-state propagation through all four approximate compiled-phase FPAA stages reports joint trace distance, the full reduced-A trace distance, and output TVD; all remain below the requested final error.

over the Ref. [4] instantiation is therefore its smaller additional working register.

Preparing a mixed Gibbs state of a noncommuting quantum Hamiltonian is distinct. Detailed-balance and KMS-Lindbladian samplers address it under Hamiltonian and jump-operator access [25]–[27]; their costs depend on Lindbladian simulation and mixing, ours on classical-chain transition oracles, adjacent QSAMPLE overlaps, and Szegedy phase gaps, so they are not direct resource competitors.

VII. DISCUSSION AND LIMITATIONS

The contribution is a one-signal-qubit GQSP realization of adjustable stationary-state phases with higher-rank intertwining, phase-sensitive clean-column analysis, and exact schedule accounting. Relative to Ref. [4], it reduces signal workspace; the improved $p_{\min}$-dependence is asymptotic only when $p_{\min} = o(1)$.

The guarantees assume controllable transition circuits, certified overlap and phase-gap bounds, and an exact initial QSAMPLE; gap certification and oracle arithmetic may dominate. The numerical reference fixes a pointwise unitary completion but synthesizes neither the complementary GQSP polynomial nor finite-precision angles, whose costs fault-tolerant estimates must add. Future work includes removing the promises, jointly optimizing schedule and transport, and extending the compiler to nonreversible chains [28].

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